EC303 — Signals and System

Module 3: Fourier Analysis, Frequency Response & Signal Space

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Authoritative Syllabus-Aligned Notes

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1. Eigenfunctions and Periodic Inputs to LSI Systems

1.1 Eigenfunctions of Continuous-Time LSI Systems

Definition: Eigenfunction of an LSI System

An input signal $x(t)$ is termed an **eigenfunction** of an LSI system $\mathcal{T}$ if the resulting output $y(t)$ is simply a scaled replica of the same input signal:

$$y(t) = \mathcal{T}\{x(t)\} = \lambda \cdot x(t) \tag{1}$$

where the complex scalar $\lambda \in \mathbb{C}$ is the corresponding **eigenvalue**.

Proof that Complex Exponentials are Eigenfunctions: Let the input to a continuous-time LSI system with impulse response $h(t)$ be the complex exponential $x(t) = e^{st}$, where $s = \sigma + j\omega \in \mathbb{C}$. By the convolution integral:

$$y(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau = \int_{-\infty}^{\infty} h(\tau)e^{s(t-\tau)}d\tau \tag{2}$$

$$= e^{st}\left[\int_{-\infty}^{\infty} h(\tau)e^{-s\tau}d\tau\right] \tag{3}$$

Notice that the bracketed integral is purely a function of the complex frequency variable s, independent of time t. Defining the system transfer function $H(s) \triangleq \int_{-\infty}^{\infty} h(\tau)e^{-s\tau}d\tau$:

$$y(t) = H(s) \cdot e^{st} \tag{4}$$

For purely imaginary frequencies $s = j\omega$, the input is the harmonic phasor $x(t) = e^{j\omega t}$, yielding:

$$y(t) = H(j\omega) \cdot e^{j\omega t} \tag{5}$$

Thus, **complex exponentials are eigenfunctions of LSI systems**, and the corresponding eigenvalue is the system's **Frequency Response** $H(j\omega)$.

1.2 Response to Periodic and Semi-Periodic Inputs

By the principle of superposition, if a periodic input is decomposed into a sum of harmonically related complex exponentials:

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} \tag{6}$$

The steady-state response of the LSI system is directly obtained by scaling each harmonic component by the frequency response evaluated at that specific harmonic:

$$y(t) = \sum_{k=-\infty}^{\infty} c_k H(jk\omega_0) e^{jk\omega_0 t} \tag{7}$$

Semi-Periodic Inputs: A semi-periodic signal is one that is periodic for $t \geq 0$ but zero for $t < 0$, represented as $x(t) = p(t)u(t)$, where $p(t+T_0) = p(t)$. When applied to an LSI system:

- The total response consists of a **transient response** $y_{tr}(t)$ and a **steady-state response** $y_{ss}(t)$:
$$y(t) = y_{tr}(t) + y_{ss}(t) \tag{8}$$
- For a BIBO stable system, the transient response decays exponentially to zero ($\lim_{t\to\infty} y_{tr}(t) = 0$), and the output asymptotically converges to the purely periodic steady-state response $y_{ss}(t)$.

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2. The Frequency Response $H(j\omega)$

2.1 Definition and Relation to Impulse Response

Definition: Frequency Response

The **Frequency Response** $H(j\omega)$ of a continuous-time LSI system is the Continuous-Time Fourier Transform (CTFT) of its unit impulse response $h(t)$:

$$H(j\omega) \triangleq \mathcal{F}\{h(t)\} = \int_{-\infty}^{\infty} h(t)e^{-j\omega t}dt \tag{9}$$

Conversely, the impulse response is recovered via the Inverse Fourier Transform:

$$h(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} H(j\omega)e^{j\omega t}d\omega \tag{10}$$

2.2 Magnitude Response and Phase Response

In polar form, $H(j\omega)$ is expressed as:

$$H(j\omega) = |H(j\omega)|e^{j\angle H(j\omega)} \tag{11}$$

where:

- $|H(j\omega)|$ is the **Magnitude Response** (or amplitude gain) of the system at frequency ω.
- $\angle H(j\omega) = \arg[H(j\omega)]$ is the **Phase Response** (phase shift) imparted to the input phasor.
- **Group Delay** $\tau_g(\omega)$**:** The time delay experienced by a narrow-band signal envelope centered at frequency ω:

$$\tau_g(\omega) \triangleq -\frac{d}{d\omega}\angle H(j\omega) \tag{12}$$

Symmetry Properties for Real Impulse Response $h(t) \in \mathbb{R}$**:**

1. Conjugate symmetry: $H(-j\omega) = H^*(j\omega)$.
2. Even magnitude symmetry: $|H(-j\omega)| = |H(j\omega)|$.
3. Odd phase symmetry: $\angle H(-j\omega) = -\angle H(j\omega)$.

Sinusoidal Steady-State Response

If the input to an LSI system is a real sinusoid $x(t) = A\cos(\omega_0 t + \theta)$, the steady-state output is:

$$y(t) = A|H(j\omega_0)|\cos\left(\omega_0 t + \theta + \angle H(j\omega_0)\right) \tag{13}$$

The frequency remains completely unchanged; the system only scales the amplitude by $|H(j\omega_0)|$ and shifts the phase by $\angle H(j\omega_0)$.

3. Continuous-Time Fourier Series (CTFS)

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3.1 Dirichlet Conditions for Convergence

A periodic signal $x(t)$ with fundamental period T_0 and fundamental frequency $\omega_0 = 2\pi/T_0$ can be expanded in a Fourier series if it satisfies the **Dirichlet Conditions**:

1. **Absolute Integrability:** Over any single period T_0, $\int_{T_0} |x(t)|dt < \infty$.
2. **Bounded Variation:** $x(t)$ has a finite number of local maxima and minima within any single period.
3. **Finite Discontinuities:** $x(t)$ has a finite number of discontinuities within any period, and each discontinuity is finite (bounded jump).

At a point of finite discontinuity t_0, the Fourier series converges to the arithmetic mean of the left- and right-hand limits: $\frac{x(t_0^+)+x(t_0^-)}{2}$.

3.2 Three Formulations of Fourier Series

Representation	Synthesis Equation	Analysis / Coefficient Equations
Trigonometric Fourier Series (TFS)	$x(t) = a_0 + \sum_{n=1}^{\infty}[a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$	$a_0 = \frac{1}{T_0}\int_{T_0} x(t)dt$, $a_n = \frac{2}{T_0}\int_{T_0} x(t)\cos(n\omega_0 t)dt$, $b_n = \frac{2}{T_0}\int_{T_0} x(t)\sin(n\omega_0 t)dt$
Compact / Polar Form	$x(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n)$	$C_0 = a_0$, $C_n = \sqrt{a_n^2 + b_n^2}$, $\theta_n = -\tan^{-1}(b_n/a_n)$
Exponential / Complex Form (EFS)	$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$	$c_k = \frac{1}{T_0}\int_{T_0} x(t)e^{-jk\omega_0 t}dt$

3.3 Inter-Relationship Between TFS and EFS Coefficients

$$c_0 = a_0 = C_0 \tag{14}$$

$$c_k = \frac{a_k - jb_k}{2} = \frac{C_k}{2}e^{j\theta_k}, \qquad c_{-k} = c_k^* = \frac{a_k + jb_k}{2} = \frac{C_k}{2}e^{-j\theta_k} \tag{15}$$

$$a_k = c_k + c_{-k} = 2\text{Re}\{c_k\}, \qquad b_k = j(c_k - c_{-k}) = -2\text{Im}\{c_k\} \tag{16}$$

3.4 Gibbs Phenomenon

When a discontinuous signal (such as a square wave) is approximated by a truncated Fourier series with N harmonics, an oscillatory overshoot occurs in the immediate vicinity of the discontinuity. As $N \to \infty$, the overshoot does not disappear; its width narrows to zero, but its peak amplitude approaches a fixed constant:

$$\text{Overshoot Percentage} = \frac{\frac{2}{\pi}\int_0^{\pi}\frac{\sin x}{x}dx - 1}{1} \approx 8.95\% \approx 9\% \text{ of the jump discontinuity} \tag{17}$$

4. Continuous-Time Fourier Transform (CTFT)

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4.1 Transition from Fourier Series to Fourier Transform

An aperiodic signal $x(t)$ can be viewed as the limiting case of a periodic signal as the period $T_0 \to \infty$. In this limit:

- The fundamental frequency $\omega_0 = 2\pi/T_0 \to d\omega$ becomes an infinitesimal differential.
- The discrete harmonic frequencies $k\omega_0 \to \omega$ merge into a continuous frequency continuum.
- The discrete spectrum $T_0 c_k$ transforms into the continuous spectral density $X(j\omega)$.

The Continuous-Time Fourier Transform (CTFT) Pair

$$\textbf{Analysis (Forward CTFT):} \quad X(j\omega) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt \tag{18}$$

$$\textbf{Synthesis (Inverse CTFT):} \quad x(t) = \mathcal{F}^{-1}\{X(j\omega)\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega \tag{19}$$

4.2 Properties of CTFT

Property	Time Domain $x(t)$	Frequency Domain $X(j\omega)$
Linearity	$ax_1(t) + bx_2(t)$	$aX_1(j\omega) + bX_2(j\omega)$
Time Shifting	$x(t - t_0)$	$X(j\omega)e^{-j\omega t_0}$
Frequency Shifting (Modulation)	$x(t)e^{j\omega_0 t}$	$X(j(\omega - \omega_0))$
Time Scaling	$x(at) \quad (a \neq 0)$	$\frac{1}{\lvert a\rvert}X\left(j\frac{\omega}{a}\right)$
Time Reversal	$x(-t)$	$X(-j\omega) = X^*(j\omega) \quad (\text{if } x(t) \in \mathbb{R})$
Duality	$X(jt)$	$2\pi x(-\omega)$
Time Differentiation	$\frac{d^n x(t)}{dt^n}$	$(j\omega)^n X(j\omega)$
Time Integration	$\int_{-\infty}^{t} x(\tau)d\tau$	$\frac{1}{j\omega}X(j\omega) + \pi X(0)\delta(\omega)$
Time Convolution	$x_1(t) * x_2(t)$	$X_1(j\omega) \cdot X_2(j\omega)$
Frequency Convolution (Multiplication)	$x_1(t) \cdot x_2(t)$	$\frac{1}{2\pi}[X_1(j\omega) * X_2(j\omega)]$

4.3 Fourier Domain Duality

The mathematical symmetry between the analysis and synthesis integrals leads to the **Duality Property**:

$$\text{If } \mathcal{F}\{x(t)\} = X(j\omega), \quad \text{then } \mathcal{F}\{X(jt)\} = 2\pi x(-\omega) \tag{20}$$

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Example: Since a rectangular pulse in time transforms into a sinc function in frequency:

$$\text{rect}\left(\frac{t}{\tau}\right) \iff \tau\text{sinc}\left(\frac{\omega\tau}{2\pi}\right) \tag{21}$$

By duality, a sinc pulse in time transforms into an ideal rectangular spectrum in frequency:

$$W\text{sinc}\left(\frac{Wt}{2\pi}\right) \iff 2\pi \cdot \text{rect}\left(\frac{\omega}{W}\right) \tag{22}$$

This establishes that ideal bandlimited filtering in frequency corresponds to infinite sinc impulse responses in time.

5. Parseval's Theorem and Energy Spectral Density

Parseval's Energy Theorem

For Continuous-Time Fourier Transform: Total signal energy E calculated in the time domain equals the energy integrated across all frequencies:

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega = \int_{-\infty}^{\infty} |X(f)|^2 df \tag{23}$$

where $f = \omega/2\pi$ is frequency in Hertz.
For Periodic Signals (CTFS): The average power P equals the sum of the average powers in all individual harmonic components:

$$P = \frac{1}{T_0}\int_{T_0} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |c_k|^2 = C_0^2 + \frac{1}{2}\sum_{n=1}^{\infty} C_n^2 \tag{24}$$

Energy Spectral Density (ESD): The quantity $S_{xx}(\omega) \triangleq |X(j\omega)|^2$ represents the distribution of energy per unit frequency (in Joules/(rad/s)). When a signal passes through an LSI system:

$$Y(j\omega) = X(j\omega)H(j\omega) \implies S_{yy}(\omega) = |Y(j\omega)|^2 = |H(j\omega)|^2 S_{xx}(\omega) \tag{25}$$

6. Discrete-Time Fourier Transform (DTFT) & Discrete Fourier Transform (DFT)

6.1 Discrete-Time Fourier Transform (DTFT)

The DTFT applies to infinite-length discrete-time sequences $x[n]$.

Definition: DTFT and Inverse DTFT

$$\textbf{Forward DTFT (Analysis):} \quad X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n} \tag{26}$$

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$$\textbf{Inverse DTFT (Synthesis):} \quad x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\Omega})e^{j\Omega n}d\Omega \tag{27}$$

Fundamental Property: 2π-Periodicity in Frequency: Since $e^{-j(\Omega+2\pi)n} = e^{-j\Omega n}e^{-j2\pi n} = e^{-j\Omega n}\cdot 1$:

$$X(e^{j(\Omega+2\pi)}) = X(e^{j\Omega}) \tag{28}$$

All distinct frequency information is fully contained within any interval of length 2π, conventionally chosen as $[-\pi, \pi]$ or $[0, 2\pi]$.

6.2 Discrete Fourier Transform (DFT)

The DFT operates on finite-duration discrete-time sequences of length N, producing an N-point discrete frequency spectrum suitable for digital computer computation.

Definition: Discrete Fourier Transform (DFT) Pair

Let $x[n]$ be an N-point sequence defined for $n = 0, 1, \dots, N-1$.

$$\textbf{DFT (Analysis):} \quad X[k] = \sum_{n=0}^{N-1} x[n]W_N^{kn}, \qquad k = 0, 1, \dots, N-1 \tag{29}$$

$$\textbf{IDFT (Synthesis):} \quad x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]W_N^{-kn}, \qquad n = 0, 1, \dots, N-1 \tag{30}$$

where the **Twiddle Factor** is $W_N \triangleq e^{-j\frac{2\pi}{N}}$.

6.3 Relation Between DTFT and DFT

The DFT values $X[k]$ are exact samples of the continuous DTFT $X(e^{j\Omega})$ evaluated at N uniformly spaced points along the unit circle:

$$X[k] = X(e^{j\Omega})\big|_{\Omega=\frac{2\pi k}{N}}, \qquad k = 0, 1, \dots, N-1 \tag{31}$$

7. Signal Space and Orthogonal Bases

7.1 The Signal Space Concept (Hilbert Space L^2)

Signals can be viewed as vectors residing in an infinite-dimensional vector space known as a **Hilbert Space** $\mathcal{H} = L^2[a, b]$. The geometric concepts of length, distance, and angle in Euclidean space generalize directly to signals:

- **Inner Product $\langle x, y\rangle$:** The inner product of two signals $x(t)$ and $y(t)$ over interval $[t_1, t_2]$ is:

$$\langle x, y\rangle \triangleq \int_{t_1}^{t_2} x(t)y^*(t)dt \tag{32}$$

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- **Signal Norm (Magnitude/Length):** The norm $\|x\|$ is defined as:

$$\|x\| \triangleq \sqrt{\langle x, x\rangle} = \left[\int_{t_1}^{t_2} |x(t)|^2 dt\right]^{1/2} = \sqrt{E_x} \tag{33}$$

  where E_x is the total energy of the signal.

- **Metric (Distance between Signals):** $d(x, y) = \|x - y\| = \sqrt{\int_{t_1}^{t_2} |x(t) - y(t)|^2 dt}$.

7.2 Orthogonality and Orthonormal Bases

Definition: Orthogonal and Orthonormal Signal Sets

A set of basis functions $\{\phi_1(t), \phi_2(t), \ldots, \phi_N(t)\}$ is:

1. **Orthogonal** if the inner product of distinct basis functions is strictly zero:

$$\langle \phi_i, \phi_j\rangle = \int_{t_1}^{t_2} \phi_i(t)\phi_j^*(t)dt = \begin{cases} 0, & i \neq j \\ E_i \neq 0, & i = j \end{cases} \tag{34}$$

2. **Orthonormal** if it is orthogonal and every basis function has unit energy ($E_i = 1$):

$$\langle \phi_i, \phi_j\rangle = \delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases} \tag{35}$$

   where δ_{ij} is the Kronecker delta.

7.3 Generalized Fourier Series Expansion

Any arbitrary signal $x(t) \in \mathcal{H}$ can be represented as a linear combination of orthonormal basis functions:

$$x(t) = \sum_{k=1}^{\infty} c_k \phi_k(t) \tag{36}$$

Taking the inner product of both sides with $\phi_m(t)$:

$$\langle x, \phi_m\rangle = \left\langle \sum_{k=1}^{\infty} c_k\phi_k, \phi_m \right\rangle = \sum_{k=1}^{\infty} c_k\langle \phi_k, \phi_m\rangle = \sum_{k=1}^{\infty} c_k\delta_{km} = c_m \tag{37}$$

Hence, the projection coefficients are given by the projection integral:

$$c_k = \langle x, \phi_k\rangle = \int_{t_1}^{t_2} x(t)\phi_k^*(t)dt \tag{38}$$

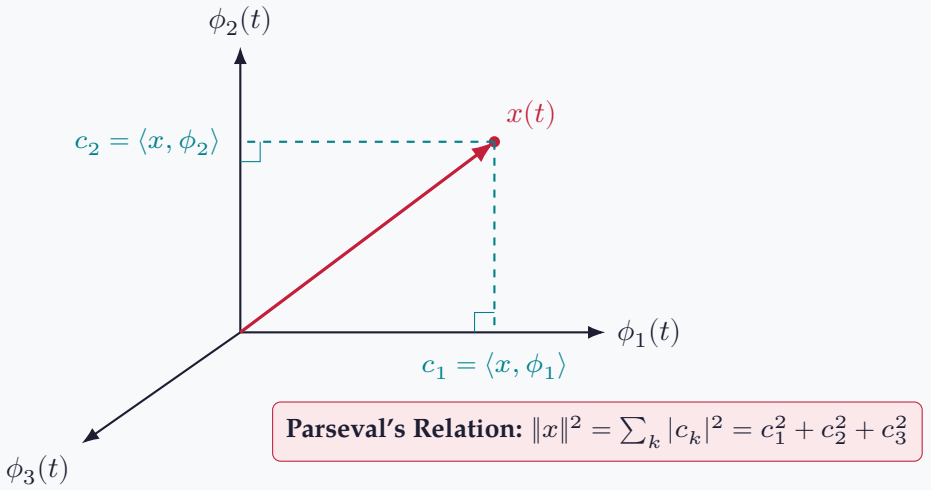
Generalized Parseval's Identity: If the basis $\{\phi_k(t)\}$ is complete:

$$\|x\|^2 = \int_{t_1}^{t_2} |x(t)|^2 dt = \sum_{k=1}^{\infty} |c_k|^2 \tag{39}$$

The continuous-time Exponential Fourier Series is simply a special case where the orthonormal basis is $\phi_k(t) = \frac{1}{\sqrt{T_0}} e^{jk\omega_0 t}$ over interval $[0, T_0]$.

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TikZ Illustration: Signal Projection onto Orthogonal Basis Axes



Parseval's Relation: $\|x\|^2 = \sum_k |c_k|^2 = c_1^2 + c_2^2 + c_3^2$

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Quick Revision & Examination Compendium

Module 3: Fourier Analysis, Frequency Response & Signal Space

Key Formulas at a Glance

Transform / Theorem	Definitive Mathematical Formula
Frequency Response $H(j\omega)$	$H(j\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t}dt = \lvert H(j\omega)\rvert e^{j\angle H(j\omega)}$
Group Delay $\tau_g(\omega)$	$\tau_g(\omega) = -\frac{d}{d\omega}\angle H(j\omega)$
Exponential Fourier Series	$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}, \quad c_k = \frac{1}{T_0}\int_{T_0} x(t)e^{-jk\omega_0 t}dt$
CTFT Analysis / Synthesis	$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt, \quad x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega$
CTFT Duality Property	$\mathcal{F}\{X(jt)\} = 2\pi x(-\omega)$
CTFT Convolution	$\mathcal{F}\{x_1(t) * x_2(t)\} = X_1(j\omega)\cdot X_2(j\omega)$
CTFT Multiplication	$\mathcal{F}\{x_1(t)\cdot x_2(t)\} = \frac{1}{2\pi}[X_1(j\omega) * X_2(j\omega)]$
Parseval's Theorem (CTFT)	$\int_{-\infty}^{\infty}\lvert x(t)\rvert^2 dt = \frac{1}{2\pi}\int_{-\infty}^{\infty}\lvert X(j\omega)\rvert^2 d\omega$
DTFT Analysis / Synthesis	$X(e^{j\Omega}) = \sum_n x[n]e^{-j\Omega n}, \quad x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\Omega})e^{j\Omega n}d\Omega$
DFT Analysis / Synthesis	$X[k] = \sum_{n=0}^{N-1} x[n]W_N^{kn}, \quad x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]W_N^{-kn}$
Orthogonality Condition	$\langle\phi_i,\phi_j\rangle = \int \phi_i(t)\phi_j^*(t)dt = \delta_{ij}\lVert\phi_i\rVert^2$

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One-Line Definitions for Rapid Review

Term	Definitive Technical Meaning
Eigenfunction	A signal that when fed into an LSI system produces an output that is an exact scalar multiple of itself.
Frequency Response	The Fourier transform of the unit impulse response, expressing gain and phase shift vs frequency.
Group Delay	The negative derivative of phase with respect to angular frequency, indicating envelope propagation delay.
Dirichlet Conditions	Sufficient mathematical conditions (integrability, bounded extrema, finite jumps) for Fourier series convergence.
Gibbs Phenomenon	The persistent $\approx 9\%$ overshoot near step discontinuities in truncated Fourier series approximations.
Duality Property	The mathematical equivalence between time and frequency roles arising from symmetrical Fourier integrals.
Energy Spectral Density	The distribution of total signal energy as a function of continuous frequency, defined as $\|X(j\omega)\|^2$.
DTFT Periodicity	The inherent 2π-periodicity of the discrete-time Fourier transform due to discrete sample spacing.
Twiddle Factor	The complex root of unity $W_N = e^{-j2\pi/N}$ utilized in discrete Fourier transform calculations.
Hilbert Space	A complete vector space equipped with an inner product that defines length, distance, and angle for signals.

Frequently Asked / Viva Questions

Q1. Why are complex sinusoids $e^{j\omega t}$ eigenfunctions of continuous-time LSI systems?

When $e^{j\omega t}$ passes through an LSI system with impulse response $h(t)$, convolution gives $y(t) = \int h(\tau)e^{j\omega(t-\tau)}d\tau = e^{j\omega t}\int h(\tau)e^{-j\omega\tau}d\tau = H(j\omega)e^{j\omega t}$. The waveform shape is perfectly preserved; only its complex amplitude is scaled by eigenvalue $H(j\omega)$.

Q2. What is the physical meaning of negative frequency $-\omega$ in Fourier analysis?

A complex phasor $e^{j\omega t}$ represents counter-clockwise rotation in the complex plane at angular speed ω. A negative frequency $e^{-j\omega t}$ represents clockwise rotation at the same speed. In real sinusoids $\cos(\omega t) = \frac{e^{j\omega t}+e^{-j\omega t}}{2}$, positive and negative frequency phasors combine symmetrically to cancel imaginary components.

Q3. State the relationship between time-domain convolution and frequency-domain multiplication.

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Frequently Asked / Viva Questions (contd.)

Convolution in the time domain corresponds to pointwise multiplication in the frequency domain: $\mathcal{F}\{x(t) * h(t)\} = X(j\omega) \cdot H(j\omega)$. Conversely, multiplication in the time domain corresponds to convolution in the frequency domain: $\mathcal{F}\{x(t) \cdot p(t)\} = \frac{1}{2\pi}[X(j\omega) * P(j\omega)]$.

Q4. What causes the Gibbs phenomenon and can it be eliminated by increasing harmonics?

Gibbs phenomenon is caused by non-uniform convergence of the Fourier series at jump discontinuities. Increasing the number of harmonics $N \to \infty$ narrows the spatial width of the ripple toward zero, but the peak amplitude remains fixed at approximately 8.95% of the jump size. It can only be eliminated by windowing/smoothing techniques.

Q5. How does the duality property allow you to find the Fourier transform of a sinc function?

Since a rectangular pulse in time $\text{rect}(t/\tau)$ transforms into $\tau\text{sinc}(\omega\tau/2\pi)$, applying duality ($\mathcal{F}\{X(jt)\} = 2\pi x(-\omega)$) reveals that a sinc pulse in time transforms into an ideal rectangular band in frequency.

Q6. Why is the DTFT periodic with period 2π while the CTFT is non-periodic?

In discrete time, the sampling index n is an integer. Thus, $e^{-j(\Omega+2\pi)n} = e^{-j\Omega n}e^{-j2\pi n} = e^{-j\Omega n} \cdot 1$. The basis functions themselves repeat every 2π, forcing the DTFT $X(e^{j\Omega})$ to be strictly periodic with period 2π.

Q7. What is the key difference between DTFT and DFT?

DTFT operates on discrete sequences and yields a continuous, 2π-periodic spectrum $X(e^{j\Omega})$. DFT operates on a finite sequence of N points and yields N discrete frequency samples $X[k]$, making DFT directly computable on digital hardware.

Q8. What is the condition for two signals to be orthogonal over an interval $[t_1, t_2]$?

Two signals $x_1(t)$ and $x_2(t)$ are orthogonal if their inner product is zero: $\langle x_1, x_2\rangle = \int_{t_1}^{t_2} x_1(t)x_2^*(t)dt = 0$.

Q9. State Parseval's relation for energy signals and explain its conservation principle.

$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$. It establishes that energy is conserved under the Fourier transformation: total energy computed across the entire time domain is identical to total energy integrated across all spectral components.

Q10. What is group delay and what condition yields zero phase distortion?

Group delay is $\tau_g(\omega) = -d\angle H(j\omega)/d\omega$. A system introduces zero phase distortion if its group delay is constant for all frequencies ($\tau_g(\omega) = t_d$), which requires a strictly linear phase response: $\angle H(j\omega) = -\omega t_d$.

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Summary Matrix: Comparison of Fourier Representations

Transform	Time Domain x	Frequency Domain X	Key Characteristics
CTFS	**Continuous, Periodic** (T_0)	**Discrete, Aperiodic** $(k\omega_0)$	Harmonics at integer multiples of ω_0.
CTFT	**Continuous, Aperiodic**	**Continuous, Aperiodic** (ω)	Continuous spectrum density; energy signals.
DTFS	**Discrete, Periodic** (N)	**Discrete, Periodic** (N)	Finite set of N unique complex amplitudes.
DTFT	**Discrete, Aperiodic**	**Continuous, Periodic** (2π)	Inherent 2π periodic frequency spectrum.
DFT	**Discrete, Finite** (N)	**Discrete, Finite** (N)	Uniform unit circle samples of DTFT; computable.

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